Seven-color two-frugal coloring for planar graphs of maximum degree four

Bostjan Bresar, Premysl Holub, Marko Jakovac, Petra Melicharova

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Editorial note. Entire original mathematical body, all native coloring diagrams and embedded bibliography retained; authors own mathematical correctness. No proof repair, additional hypothesis, case completion or independent certification.

Editorial note. The full original proof section is selected, including native narrative outer proof and every claim/recoloring/discharging component. No artificial proof environment is introduced.

Editorial note. Degree-four bound seven and degree-five bound ten preserved exactly; intermediate forbidden configurations and open degree-six/seven questions are not separately counted.

Editorial note. Revision r002 supplies original forloop.sty as a local asset, with upstream LGPL README, original corresponding source and provenance. Package code and author mathematical source unchanged; failed r001 retained and no global installation performed.

Allowed standard prerequisites (editorial).

- Planar graph coloring, neighborhood frugality, reducible configurations and discharging. Native finite simple graph convention, proper two-frugal definition, seven-versus-ten palette bounds, every minimal-counterexample claim and exact discharging rules retained.. All original inline TikZ coloring diagrams and complete proof sections supplied; no recoloring case filled in or independently checked.

1 Introduction

A 2-*distance coloring* of a graph G is a coloring of the vertices of G such that vertices at distance at most two obtain distinct colors. Alternatively, it is a proper coloring of the square G^2 of G, which is the graph obtained from G by adding edges between vertices at distance 2. The 2-*distance chromatic number* of G is the minimum number of colors in a 2-distance coloring of G, and is denoted by $\chi^2(G)$. The concept of d-distance coloring, where d is a positive integer, was introduced in [4] and surveyed in [5].

A famous conjecture due to Wegner [7] concerns the 2-distance chromatic number of planar graphs:

Conjecture 1.1 (Wegner [7])**.** *If G is planar with maximum degree Δ, then*

$$\chi^2(G) \leq \begin{cases} 7, & \Delta \leq 3, \\ \Delta + 5, & 4 \leq \Delta \leq 7, \\ \left\lfloor \frac{3\Delta}{2} \right\rfloor + 1, & \Delta \geq 8. \end{cases}$$

The conjecture has been proven for $\Delta = 3$ (see Hartke et al. [2] and Thomassen [6]), but remains open in general. Addressing weaker versions of the conjecture may help solving the original problem. Among coloring invariants that lie between coloring and 2-distance coloring, one can investigate a frugal coloring, introduced in 1997 by Hind, Molloy and Reed [3], and defined as follows. For a positive integer t, a proper vertex coloring of a graph G is *t-frugal* if no color appears more than t times in the neighborhood of any vertex in G. The *t-frugal chromatic number* of G is the minimum k for which G admits a t-frugal coloring with k colors, and we denote it by $\chi_t^f(G)$. Clearly, the condition of t-frugal coloring is weaker than that of 2-distance coloring, and its strength depends on t as described in the following chain of inequalities:

$$\chi(G) = \chi_\Delta^f(G) \leq \ldots \leq \chi_2^f(G) \leq \chi_1^f(G) = \chi^2(G).$$

In this paper, we consider a weaker version of Wegner's conjecture by using t-frugal colorings. Such a result would be of independent interest, but may also lead to an approach to resolve the conjecture itself. The following problem arises naturally:

Problem 1.2. *Prove that for every planar graph G with maximum degree $\Delta \in \{4, 5, 6, 7\}$,*

$$\chi_2^f(G) \leq \Delta + 5.$$

Relaxations similar to frugal coloring have been fruitful: Cho, Choi and Lidický [1] considered a relaxation of Wegner's conjecture for graphs with maximum degree 4. They proved that every planar graph has a proper 9-coloring such that at most one color is allowed to be repeated in the neighborhood of a vertex of degree 4 (but nowhere else). Note that the condition of 2-frugal coloring is slightly weaker (since we allow that for every vertex a color may be repeated in its neighborhood, and for vertices of degree 4 we even allow that two colors are repeated). Hence, the result from [1] implies that $\chi_2^f(G) \leq 9$ holds for every planar graph G with maximum degree 4. Here we improve this upper bound by proving the following statement.

Theorem 1.3. *For every planar graph G with $\Delta(G) \leq 4$,*

$$\chi_2^f(G) \leq 7.$$

For planar graphs with $\Delta \geq 5$, Problem 1.2 is widely open. In this paper, we resolve Problem 1.2 for the case $\Delta = 5$.

Theorem 1.4. *For every planar graph G with $\Delta(G) \leq 5$,*

$$\chi_2^f(G) \leq 10.$$

Section 2 is devoted to the proof of Theorem 1.3, and Section 3 contains the proof of Theorem 1.4. In the next paragraph we give some basic definitions needed throughout the paper.

In this paper, we consider finite, simple graphs. Given a graph G and a vertex $v \in V(G)$, the *neighborhood* of v in G is $N_G(v) = \{u \in V(G) : uv \in E(G)\}$ and its *closed neighborhood* is $N_G[v] = N_G(v) \cup \{v\}$. The *degree*, $\deg_G(v)$, of v in G is $|N_G(v)|$, while $\delta(G)$ and $\Delta(G)$ are the minimum, resp. the maximum degree of vertices in G. We denote the order of a graph G by $n(G)$. For convenience, let $[n] = \{1, \ldots, n\}$.

2 Proof of Theorem 1.3

The proof is by a contradiction, assuming that there exists a counter-example. Let G be a minimum counter-example to the statement of the theorem with respect to the order of G. Thus, G is a planar graph with $\Delta(G) \leq 4$ such that $\chi_2^f(G) > 7$, while every planar graph G' with order $n(G') < n(G)$ has $\chi_2^f(G') \leq 7$. First, we will prove two claims, which restrict the structure of the counter-example G, and then, by using a simple discharging argument, we will obtain a contradiction with G being planar.

The following proof technique will be used in all proofs of our claims. A set X, where $X \subset V(G)$, will be removed from G so that $G' = G - X$ has smaller order and thus admits a 2-frugal coloring c. Our goal is to extend c from G' to G, and while doing so, some of the colors cannot be used, and we call them forbidden colors. If $x \in X$, then the colors $\{c(y) : y \in N_G(x) \cap V(G')\}$ are forbidden for x, since we want c to be a proper coloring of G. In addition, for every $y \in N_G(x)$, any eventual color $\alpha \in [7]$ that appears twice in $N_G(y) \cap V(G')$ is forbidden for x, since assigning $c(x) = \alpha$ would result in the neighborhood of y in G having three vertices with color α. Any color that is not forbidden for x is called a free color for x. Now, assigning a free color to every vertex of X implies that each vertex y in G' will be properly colored with respect to 2-frugal coloring condition; notably neighbors of y in G will receive a different color from $c(y)$ and each color will appear at most two times in $N_G(y)$. However, to ensure that each vertex x in X will be properly colored with respect to 2-frugal coloring condition we need to use a free color for every such vertex and additionally prevent "collisions" between vertices in X. Thus, while coloring vertices of X one at a time, the number of free colors of not yet colored vertices in X may need to be further reduced.

Claim 1. $\delta(G) \geq 4$.

Proof. Suppose x is a vertex of degree 3 in G and let $N_G(x) = \{x_1, x_2, x_3\}$. Let $X_i = N_G(x_i) \setminus \{x\}$, and note that $|X_i| \leq 3$ for all $i \in [3]$. Consider G' obtained from $G - x$ by possibly adding an edge between x_1 and x_2 (if $x_1x_2 \notin E(G)$). Since G' has smaller order than G and $\Delta(G') \leq 4$, there exists a 2-frugal coloring c of G' using 7 colors. Note that one can extend c to a proper coloring of G by assigning to x a color not in $\{c(x_1), c(x_2), c(x_3)\}$. Now, c is also 2-frugal, unless $c(x)$ coincides with a color that appears twice in some X_i. Note that there are at most three forbidden colors that arise from $|N_G(x)| = 3$, and for each $i \in [3]$ there is only one forbidden color that arises from $|X_i| \leq 3$ and potential repetitions of colors in $N_G(x_i)$. Altogether, we infer there are at most $3 + 3 \times 1 = 6$ forbidden colors for the vertex x. (See also Fig. 1 depicting a possible situation that can appear with the coloring in G'; note that in each neighborhood X_i there is a vertex whose color does not influence the resulting color for x and thus the colors of these vertices are omitted in the picture.) Thus, there is at least one free color for x, and so c can be extended to G, resulting in a 2-frugal coloring of G, a contradiction. □

Claim 2. *There is no triangle in G.*

Proof. Suppose to the contrary that x, y, z are the vertices of a triangle in G. By Claim 1, G is 4-regular. Hence $\deg_G(x) = \deg_G(y) = \deg_G(z) = 4$. Consider $G'' = G - \{x, y, z\}$, and let $N_G(x) \cap V(G'') = \{x_1, x_2\}$, $N_G(y) \cap V(G'') = \{y_1, y_2\}$, and $N_G(z) \cap V(G'') = \{z_1, z_2\}$. Further let $X_i = N_G(x_i) \setminus \{x\}$, $Y_i = N_G(y_i) \setminus \{y\}$ and $Z_i = N_G(z_i) \setminus \{z\}$ for all $i \in [2]$. Now let G' be obtained from graph G'' by adding the edges x_1x_2, y_1y_2 and z_1z_2 if they are not already in G (see Fig. 2). Since G' has smaller order than G and $\Delta(G') \leq 4$, there exists a 2-frugal coloring c of G' using 7 colors.

In the next step we count the number of free colors at vertices x, y and z in G. Note that $|X_i| \leq 3$ for each $i \in [2]$, which together with colors $c(x_1)$ and $c(x_2)$ adds up to at most

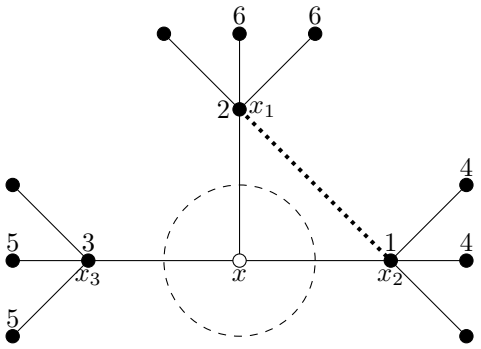


Figure 1: A possible coloring of some vertices in G' in Claim 1. Dashed circle describes the removal of x, while the dotted edge is added in G' (if it is not already in G). Note that in this example, there is exactly one free color at x.

$2\times 1+2=4$ forbidden colors for x. Since $y_1y_2 \in E(G')$ and $z_1z_2 \in E(G')$, we have $c(y_1) \neq c(y_2)$ and $c(z_1) \neq c(z_2)$, thus regardless of the color assigned to x, no three vertices in $N_G(y)$ or $N_G(z)$ will receive the same color. Hence, x indeed has at least three free colors. By symmetry, we derive that y as well as z has at least three free colors. Therefore, one can easily extend c to a 2-frugal coloring of G, a contradiction. □

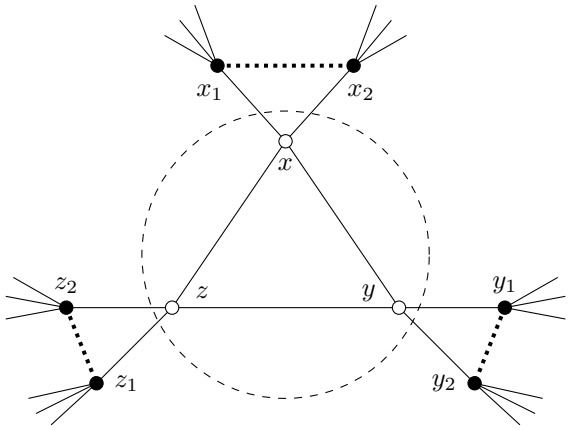


Figure 2: The situation in Claim 2. Dashed circle describes the removal of the triangle, while dotted edges are added in G' (if they are not already there in G).

We finish the proof of Theorem 1.3 by using a basic discharging argument. Letting $F(G)$ be the set of faces of G, we assign the initial charge $\mu(z)$ for each element $z \in V(G) \cup F(G)$ to be $\deg_G(z) - 4$ (note that the degree of a face is defined as the number of its boundary edges (or vertices)). By Euler's formula the initial total charge of the graph, which is the sum of all charges, equals

$$\sum_{v\in V(G)} (\deg_G(v) - 4) + \sum_{f\in F(G)} (\deg_G(f) - 4) = 2|E(G)| - 4|V(G)| + 2|E(G)| - 4|F(G)| = -8.$$

By Claims 1 and 2, $\delta(G) = 4$ and all the faces of G have lengths at least 4, we have

$$\sum_{v \in V(G)} (\deg_G(v) - 4) + \sum_{f \in F(G)} (\deg_G(f) - 4) \geq 0,$$

which is a contradiction. The proof is complete.

3 Proof of Theorem 1.4

The proof follows similar steps and uses techniques similar to, though somewhat more complex than, those in the proof of Theorem 1.3. Suppose to the contrary that there exists a counter-example to the theorem, and let G be a counter-example with minimum order. Thus, G is a planar graph with $\Delta(G) \leq 5$ such that $\chi_2^f(G) > 10$, yet every planar graph G' with order $n(G') < n(G)$ has $\chi_2^f(G') \leq 10$. In the sequel, we will prove a sequence of claims, which restrict the structure of the counter-example G. In the end, by using a discharging argument we will obtain a contradiction with G being planar.

Claim 3. $\delta(G) \geq 4$.

Proof. Suppose x is a vertex of degree 3 in G and let $N_G(x) = \{x_1, x_2, x_3\}$. Let $Y_i = N(x_i) \setminus \{x\}$, and note that $|Y_i| \leq 4$ for all $i \in [3]$. Consider G' obtained from $G - x$ by possibly adding an edge between x_1 and x_2 (if $x_1x_2 \notin V(G)$). Since G' has smaller order than G, there exists a 2-frugal coloring c of G' using 10 colors. Note that one can extend c to a proper coloring of G by assigning to x a color not in $\{c(x_1), c(x_2), c(x_3)\}$. Now, c is also 2-frugal, unless $c(x)$ coincides with a color that appears twice in some Y_i. Note that there are at most three forbidden colors that arise from $|N_G(x)| = 3$, and for each $i \in [3]$ there are most two forbidden colors that arise from $|Y_i| = 4$ and potential repetitions of colors in $N_G(x_i)$. Altogether, we infer there are at most $3 + 3 \times 2 = 9$ forbidden colors for $c(x)$ (see also Fig. 3 depicting a possible situation that can appear with the coloring in G'). Thus, there is at least one free color for x, and so c can be extended to G, resulting in a 2-frugal coloring of G (using ten colors), a contradiction. □

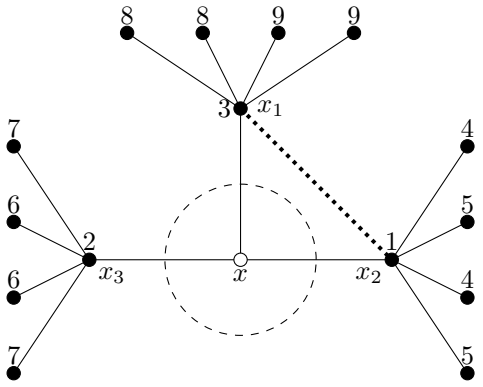


Figure 3: A possible coloring of some vertices in G' in Claim 3; there is only one free color at x.

Claim 4. *Every triangle in G has at least two vertices of degree 5 in G.*

Proof. Let x, y, z be the vertices of a triangle in G, and let us assume that $\deg_G(x) = 5$ and $\deg_G(y) = \deg_G(z) = 4$. (If the degrees of y or z in G are less than 4, then the proof follows similar lines while there are even more free colors for some vertices.) Consider $G'' = G - \{x, y, z\}$, and let $N_G(x) \cap V(G'') = \{x_1, x_2, x_3\}$, $N_G(y) \cap V(G'') = \{y_1, y_2\}$, and $N_G(z) \cap V(G'') = \{z_1, z_2\}$. Further let $X_i = N_G(x_i) \setminus \{x\}$ for all $i \in [3]$, and $Y_i = N_G(y_i) \setminus \{y\}$, $Z_i = N_G(z_i) \setminus \{z\}$ for all $i \in [2]$. Now let G' be obtained from graph G'' by adding the edges x_1x_2, y_1y_2 and z_1z_2 if necessary (that is, if these edges are not already in G; see Fig. 4). Since G' has smaller order than G and $\Delta(G) \leq 5$, there exists a 2-frugal coloring c of G' using 10 colors.

In the next step we count the number of free colors at vertices x, y and z in G. Note that $|X_i| \leq 4$ for each $i \in [3]$, which together with colors $c(x_1)$, $c(x_2)$ and $c(x_3)$ adds up to at most $2 \times 3 + 3 = 9$ forbidden colors for x. Since $y_1y_2 \in E(G')$ and $z_1z_2 \in E(G')$, we have $c(y_1) \neq c(y_2)$ and $c(z_1) \neq c(z_2)$, thus regardless of the color assigned to x, no three vertices in $N_G(y)$ or $N_G(z)$ will receive the same color. Hence, x indeed has at least one free color. We now focus on the vertex y. Note that $|Y_i| \leq 4$ for each $i \in [2]$, which together with colors $c(y_1)$ and $c(y_2)$ adds up to at most $2 \times 2 + 2 = 6$ forbidden colors. However, it is possible that either $c(x_3) = c(x_1)$ or $c(x_3) = c(x_2)$, which potentially yields an additional forbidden color for y. Thus, altogether, y has at least three free colors. Similarly, z also has at least three free colors.

Next, we color the vertices of the triangle. First we color the vertex x with a free color. Consequently the number of free colors at vertices y and z can drop by one, resulting in at least two free colors at both vertices y and z. In the next step, we color the vertex y with a free color, again potentially dropping the number of free colors at z to one. Thus, we can color z in the final step, which gives a 2-frugal coloring of G, a contradiction, which yields the statement. □

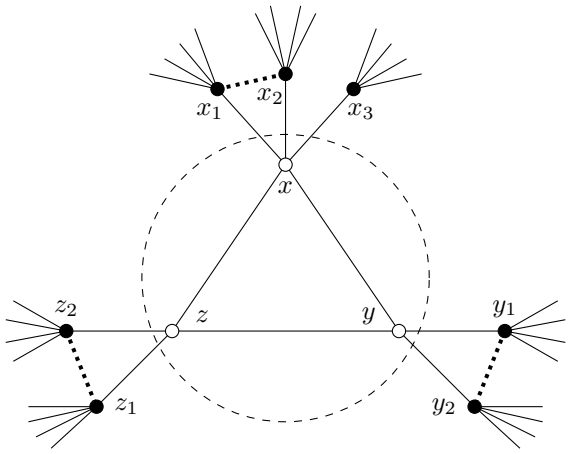


Figure 4: The situation in Claim 4. Dashed circle describes the removal of the triangle, while dotted edges are added in G' (if they were not already there in G).

Claim 5. *Every vertex of a diamond in G has degree 5 in G.*

Proof. Let x, y, z, z', where $xy \notin E(G)$, be the vertices of a diamond D in G, and suppose that one of those vertices has degree 4. We split the proof into two subcases.

First, let $\deg_G(y) = 4$ and $\deg_G(x) = \deg_G(z) = \deg_G(z') = 5$. (If the degrees of y, z or z' in G are less than 5, the proof follows similar lines while there may be even more free colors for some vertices.) Consider $G'' = G - D$, and let $N_G(x) \cap V(G'') = \{x_1, x_2, x_3\}$,

$N_G(y) \cap V(G'') = \{y_1, y_2\}$, $N_G(z) \cap V(G'') = \{z_1, z_2\}$, and $N_G(z') \cap V(G'') = \{z'_1, z'_2\}$. Further let $X_i = N_G(x_i) \setminus \{x\}$ for all $i \in [3]$, and $Y_i = N_G(y_i) \setminus \{y\}$, $Z_i = N_G(z_i) \setminus \{z\}$ and $Z'_i = N_G(z'_i) \setminus \{z'\}$ for all $i \in [2]$. Now let G' be obtained from graph G'' by adding the edges x_1x_2, y_1y_2, z_1z_2 and $z'_1z'_2$ if necessary (see Fig. 5(a)). Since G' has smaller order than G and $\Delta(G') \leq 5$, there exists a 2-frugal coloring c of G' using 10 colors.

Let us count the number of free colors at vertices x, y, z and z' in G. Note that $|X_i| \leq 4$ for each $i \in [3]$, which together with colors $c(x_1)$, $c(x_2)$ and $c(x_3)$ adds up to at most $3 \times 2 + 3 = 9$ forbidden colors at x. Since $z_1z_2 \in E(G')$ and $z'_1z'_2 \in E(G')$, we have $c(z_1) \neq c(z_2)$ and $c(z'_1) \neq c(z'_2)$, thus regardless of the color assigned to x, no three vertices in $N_G(z)$ and $N_G(z')$ will receive the same color. Hence, x indeed has at least one free color. We now focus on the vertex y. Note that $|Y_i| \leq 4$ for each $i \in [2]$, which together with colors $c(y_1)$ and $c(y_2)$ adds up to at most $2 \times 2 + 2 = 6$ forbidden colors. By the same reasoning as for the vertex x, since $c(z_1) \neq c(z_2)$ and $c(z'_1) \neq c(z'_2)$, we get no more forbidden colors for y. Hence y has four free colors. Finally, let us count the forbidden colors for z and z'. Note that $|Z_i| \leq 4$ for each $i \in [2]$, which together with colors $c(z_1)$ and $c(z_2)$ adds up to at most $2 \times 2 + 2 = 6$ forbidden colors. Since it is possible that either $c(x_3) = c(x_1)$ or $c(x_3) = c(x_2)$, this potentially yields an additional forbidden color for z. Thus, altogether, z has at least three free colors. Similarly, by symmetry, z' has at least three free colors.

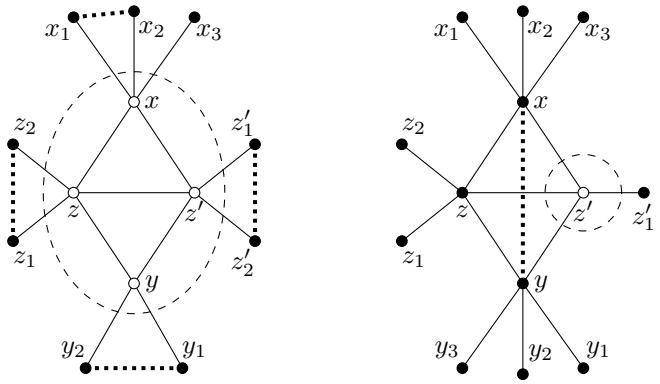
Next, we color the vertices of the diamond. First, color the vertex x with a free color. Now the number of free colors at vertices z and z' can drop by one, resulting in at least two free colors at each z and z'. Moreover, the color assigned to x can coincide with a color already used in $N_G(z)$ and/or $N_G(z')$, and therefore becomes forbidden for y. Hence, also the number of free colors at y can drop to three. In the next two steps we color the vertices z and z' in any order, which can be done since both have two free colors. This can result in two more forbidden colors for y, leaving at least one free color at y. Thus, y can also be colored, which gives a 2-frugal coloring of G, a contradiction.

In the second case, let $\deg_G(z') = 4$ and $\deg_G(x) = \deg_G(y) = \deg_G(z) = 5$. (If the degrees of x, y or z in G are less than 4, then the proof again follows similar lines while there can be even more free colors for some vertices.) Consider $G'' = G - z'$, and let $N_G(x) \cap V(G'') = \{x_1, x_2, x_3, z\}$, $N_G(y) \cap V(G'') = \{y_1, y_2, y_3, z\}$, $N_G(z) \cap V(G'') = \{z_1, z_2, x, y\}$, and $N_G(z') \cap V(G'') = \{z'_1, x, y, z\}$. Further let $X_i = N_G(x_i) \setminus \{x\}$ and $Y_i = N_G(y_i) \setminus \{y\}$ for all $i \in [3]$, $Z_i = N_G(z_i) \setminus \{z\}$ for all $i \in [2]$, and $Z'_1 = N_G(z'_1) \setminus \{z'\}$. Now let G' be obtained from graph G'' by adding the edge xy (see Fig. 5(b)). Since G' has smaller order than G and $\Delta(G') \leq 5$, there exists a 2-frugal coloring c of G' using 10 colors.

Note that the colors at x, y and z are pairwise different since they form a triangle in G', hence there are at most two vertices in $N_G(z')$ given the same color by c. Now, we count the number of free colors at vertex z'. Note that $|Z'_1| \leq 4$, which together with color $c(z'_1)$ adds up to at most $2 + 1 = 3$ forbidden colors. Also, we need to forbid the colors given to x, y and z, since they are neighbors of z' in G yielding at most six forbidden colors at z'. Additionally, it is possible that two of the vertices x_1, x_2 and x_3 received the same color in G', and also two of the vertices y_1, y_2 and y_3 received the same color in G', which potentially yields two more forbidden colors for z'. Finally, if $c(z_1) = c(z_2)$, then this yields another forbidden color for z'. Altogether, at most nine colors are forbidden at z'. Thus, c can be extended to G, resulting in a 2-frugal coloring of G, a contradiction. □

Claim 6. *There is no subgraph isomorphic to K_4 in G.*

Proof. Let x, y, z, w be the vertices of a subgraph K_4 in G, and let us assume that $\deg_G(x) = \deg_G(y) = \deg_G(z) = \deg_G(w) = 5$. (If some of the degrees of these vertices in G are less



(a) (b)

Figure 5: Two cases in the proof of Claim 5. Dashed circle describes the removal of the subgraph, while dotted edges are added in G' (if they were not already there in G); black vertices may have additional neighbors with respect to the maximum degree.

than 5, the proof follows similar lines while there are even more free colors for some vertices.) Consider $G'' = G - V(K_4)$, and let $N_G(x) \cap V(G'') = \{x_1, x_2\}$, $N_G(y) \cap V(G'') = \{y_1, y_2\}$, $N_G(z) \cap V(G'') = \{z_1, z_2\}$ and $N_G(w) \cap V(G'') = \{w_1, w_2\}$. Further let $X_i = N_G(x_i) \setminus \{x\}$, $Y_i = N_G(y_i) \setminus \{y\}$, $Z_i = N_G(z_i) \setminus \{z\}$ and $W_i = N_G(w_i) \setminus \{w\}$ for all $i \in [2]$. Now let G' be obtained from graph G'' by adding the edges x_1x_2, y_1y_2, z_1z_2 and w_1w_2 if necessary (see Fig. 6). Since G' has smaller order than G and $\Delta(G') \leq 5$, there exists a 2-frugal coloring c of G' using 10 colors.

In the next step we count the number of free colors at vertices x, y, z and w in G. Note that $|X_i| \leq 4$ for each $i \in [2]$, which together with colors $c(x_1)$ and $c(x_2)$ adds up to at most $2 \times 2 + 2 = 6$ forbidden colors for x. Thus, x has four free colors. By symmetry, the same holds true for vertices y, z and w, which all have four free colors.

Finally, we color the vertices x, y, z and w of the subgraph K_4, and this can be done in any order since all have four free colors, and since all have to receive different colors, this does not violate the 2-frugal coloring condition. This gives a 2-frugal coloring of G, a contradiction. □

Claim 7. *Every vertex of degree* 4 *in* G *belongs to at most one triangle in* G*.*

Proof. Let x be a vertex of degree 4 in G belonging to more than one triangle in G, and let x_1, x_2, x_3, x_4 denote its neighbors in G. Without loss of generality assume that $x_1x_2 \in E(G)$. Combining Claims 5 and 6, $x_2x_3 \notin E(G)$ and $x_1x_4 \notin E(G)$. Thus $x_3x_4 \in E(G)$. Let $G' = G - x$ (see Fig. 7). Since G' has smaller order that G and $\Delta(G') \leq 5$, G' has a 2-frugal coloring c of G' using 10 colors.

Since every vertex x_i, $i \in [4]$, has three neighbors in $G - N_G[x]$ by Claim 4, there are at most 8 forbidden colors for x (together with $c(x_1)$, $c(x_2)$, $c(x_3)$ and $c(x_4)$). Thus there is a free color for x, implying that c can be extended to a 2-frugal coloring of G, a contradiction. □

Claim 8. *There is no subgraph isomorphic to a* 4*-wheel in* G*.*

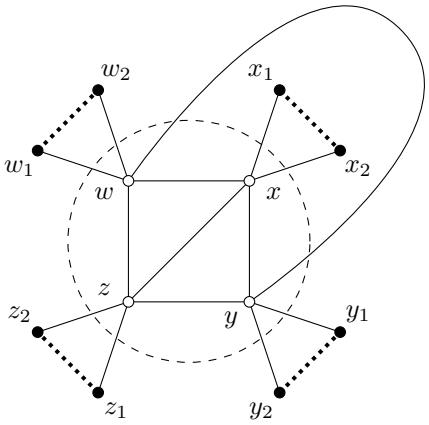


Figure 6: The situation in Claim 6. Dashed circle describes the removal of the K_4, while dotted edges are added in G' (if they were not already there in G); black vertices may have additional neighbors with respect to the maximum degree.

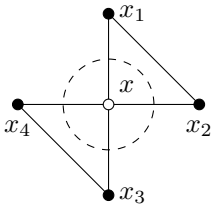


Figure 7: The situation in Claim 7. Dashed circle describes the removal of the vertex of degree 4; black vertices may have additional neighbors with respect to the maximum degree.

Proof. Let x, y, z, u, v be the vertices of a 4-wheel W in G with the central vertex x. Since each of these vertices belongs to a diamond, we infer $\deg_G(x) = \deg_G(y) = \deg_G(z) = \deg_G(u) = \deg_G(v) = 5$. Consider $G'' = G - V(W)$, and let $N_G(x) \cap V(G'') = \{x_1\}$, $N_G(y) \cap V(G'') = \{y_1, y_2\}$, $N_G(z) \cap V(G'') = \{z_1, z_2\}$, $N_G(u) \cap V(G'') = \{u_1, u_2\}$ and $N_G(v) \cap V(G'') = \{v_1, v_2\}$. Further let $X_1 = N_G(x_1) \setminus \{x\}$, and $Y_i = N_G(y_i) \setminus \{y\}$, $Z_i = N_G(z_i) \setminus \{z\}$, $U_i = N_G(u_i) \setminus \{u\}$ and $V_i = N_G(v_i) \setminus \{v\}$ for all $i \in [2]$. Now let G' be obtained from G'' by adding the edges y_1y_2, z_1z_2, u_1u_2 and v_1v_2 if necessary (see Fig. 8). Since G' has smaller order than G and $\Delta(G') \leq 5$, there exists a 2-frugal coloring c of G' using 10 colors.

Let us count the number of free colors at vertices of the 4-wheel W in G. Note that $|X_1| \leq 4$, which together with the color $c(x_1)$ adds up to at most $2 \times 1 + 1 = 3$ forbidden colors for x. Thus, x has seven free colors. For the remaining vertices of W we proceed in the same manner. Note that $|Y_i| \leq 4$ for each $i \in [2]$, which together with colors $c(y_1)$ and $c(y_2)$ adds up to at most $2 \times 2 + 2 = 6$ forbidden colors for y. Thus, y has four free colors, and the same holds true for vertices z, u and v, which all have four free colors.

Finally, we color the vertices of the 4-wheel W in the following way. Firstly, we can color the vertices y, z, u and v with pairwise different colors in any order since they all have four free

colors. This will not violate the frugal coloring condition in $N_G(x)$ even if one of those colors is the same as the color $c(x_1)$. After coloring these four vertices, the number of free colors for x can drop to three, and we can use one of those colors for x to obtain a 2-frugal coloring of G, which is a contradiction. □

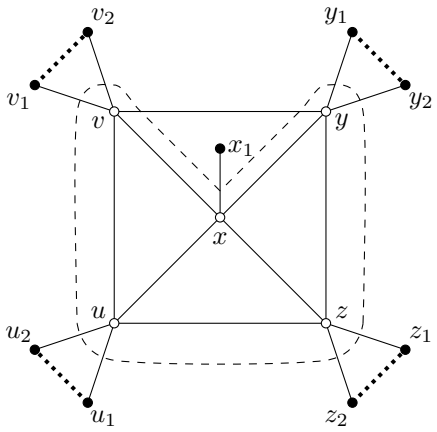


Figure 8: The situation in Claim 8. Dashed circle describes the removal of the 4-wheel, while dotted edges are added in G' (if they were not already there in G); black vertices may have additional neighbors with respect to the maximum degree.

Claim 9. *There is no vertex of degree* 5 *in* G *which belongs to exactly four triangles in* G*.*

Proof. Suppose to the contrary that x is such a vertex, and let y, z, u, v, w be its neighbors. Without loss of generality assume that $\{yz, zu, uv, vw\} \subset E(G)$ whereas $yw \notin E(G)$, which results in exactly four triangles to which x belongs. Due to Claims 6 and 8, there are no additional edges between any two vertices in $\{y, z, u, v, w\}$, and due to Claim 5 the degree of each of these vertices is 5. Let $G'' = G - N_G[x]$, and let $N_G(y) \cap V(G'') = \{y_1, y_2, y_3\}$, $N_G(z) \cap V(G'') = \{z_1, z_2\}$, $N_G(u) \cap V(G'') = \{u_1, u_2\}$, $N_G(v) \cap V(G'') = \{v_1, v_2\}$, $N_G(w) \cap V(G'') = \{w_1, w_2, w_3\}$. Furthermore, $Y_i = N_G(y_i) \setminus \{y\}$, $W_i = N_G(w_i) \setminus \{w\}$, where $i \in [3]$, and $Z_i = N_G(z_i) \setminus \{z\}$, $U_i = N_G(u_i) \setminus \{u\}$, $V_i = N_G(v_i) \setminus \{v\}$, where $i \in [2]$. Let G' be the graph obtained from G'' by adding edges $y_1y_2, z_1z_2, u_1u_2, v_1v_2, w_1w_2$ whenever they are not already in G'' (see Fig. 9). Since G' has smaller order than G and $\Delta(G') \leq 5$, G' has a 2-frugal coloring c using 10 colors.

Let us count the numbers of free colors at vertices in $V(G) \setminus V(G')$. Note that y and w each have at most $2 \times 3 + 3 = 9$ forbidden colors, whereas z, u, v each have at most $2 \times 2 + 2 = 6$ forbidden colors. For the vertex x, since $\{y_1y_2, z_1z_2, u_1u_2, v_1v_2, w_1w_2\} \subset E(G')$, the only forbidden colors appear at x when $c(y_i) = c(y_3)$ or $c(w_i) = c(w_3)$ for some $i \in [2]$; thus altogether there are at most two forbidden colors at x.

Now we extend the coloring c of G' to G. We start by coloring the vertices y and w, which each have at least one free color. Note that if y and w have a common neighbor (that is, $y_i = w_j$ for some i and j), then actually y and w each have at least two free colors. In such a case, we color y and w with two distinct colors. Otherwise, if y and w do not have a common neighbor, then color each of them with its own free color. (Note that in the latter case, it is possible that $c(y) = c(w)$.) In the rest of the proof, we consider each of these two possibilities.

First, assume that $c(y) = c(w)$. This implies possibly one more forbidden color for each of z, u, v (for z and v since y or w is its neighbor, for u since x has two neighbors y and w having the same color). Thus, z has at least 3 free colors, and after coloring it, this may decrease numbers of free colors for u and v by one. Hence u as well as v has at least two free colors. Now color u with a free color, which may again decrease the number of free colors at v by one. Thus, v has at least one free color, and we next color v. Finally, vertex x initially had at most two forbidden colors, and in its neighborhood there are at most 4 colors, which are additionally forbidden for x. Hence there are at least 4 free colors at x and we color it. Therefore G has a 2-frugal coloring using at most 10 colors, a contradiction.

Second, assume that $c(y) \neq c(w)$. If $c(w)$ is a free color for z, we assign z the color $c(w)$, which decreases the number of free colors at u by one. Hence u still has at least two free colors (indeed, from initial four free colors at u, the colors $c(y)$ and $c(w)$ are also forbidden for u). Since v is a neighbor of w, $c(w)$ is already forbidden for v so the number of free colors at v after coloring of z remains the same (at least two). Then we color u with a free color, which decreases the number of free colors for v by 1. However, at least one free color for v remains implying that we obtain a 2-frugal coloring of G, a contradiction. If $c(y)$ is a free color for v, we are in the previous situation interchanging the roles of z and v. Thus we may assume that $c(w)$ is not free for z and $c(y)$ is not free for v. Vertex u has at least 2 free colors (4 initially minus possibly $c(w)$ and $c(y)$). We color u with a free color, which decreases the numbers of free colors of z and v by one. For z (and, symmetrically, for v), there are at most 8 forbidden colors (6 initially plus $c(w)$ and $c(u)$). Color z with a free color, which decreases the number of free colors at v by one. Finally, we have a free color at v, implying that we can extend the coloring c of G' to a 2-frugal coloring of G, a contradiction. □

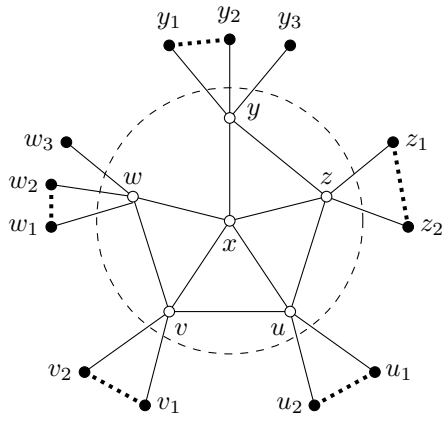


Figure 9: Situation from the proof of Claim 9. Dashed circle describes the removal of the subgraph, while dotted edges are added in G' (if they were not already there in G); black vertices may have additional neighbors with respect to the maximum degree.

Claim 10. *If a vertex of degree* 5 *is the center of a* 5*-wheel, then all its neighbors lie in exactly two triangles (of the same wheel).*

Proof. Let x, y, z, u, v, w be the vertices of a 5-wheel W in G with the central vertex x. Note that by Claim 6, W is an induced 5-wheel. Moreover, by Claim 5, every vertex from W lies in a diamond, hence $\deg_G(x) = \deg_G(y) = \deg_G(z) = \deg_G(u) = \deg_G(v) = \deg_G(w) = 5$.

Suppose to the contrary, that y is a vertex that lies in more than two triangles in G. There are two situations for this to happen, and we split the proof into two subcases correspondingly.

Firstly suppose that y lies in a triangle whose remaining two vertices (y_1 and y_2) do not belong to W. Consider the graph $G'' = G - \{x, y, z, u\}$, and let $N_G(y) \cap V(G'') = \{y_1, y_2\}$, $N_G(z) \cap V(G'') = \{z_1, z_2\}$, $N_G(u) \cap V(G'') = \{u_1, u_2\}$, $N_G(v) \cap V(G'') = \{v_1, v_2\}$ and $N_G(w) \cap V(G'') = \{w_1, w_2\}$. Further let $Y_i = N_G(y_i) \setminus \{y, y_{3-i}\}$, $Z_i = N_G(z_i) \setminus \{z\}$, $U_i = N_G(u_i) \setminus \{u\}$, $V_i = N_G(v_i) \setminus \{v\}$ and $W_i = N_G(w_i) \setminus \{w\}$ for all $i \in [2]$. Now let G' be obtained from graph G'' by also adding the edges z_1z_2 and u_1u_2 if necessary (see Figure 10(a)). Since G' has smaller order than G and $\Delta(G') \leq 5$, there exists a 2-frugal coloring c of G' using 10 colors.

Let us count the number of free colors at vertices x, y, z and u in G. Note that x is adjacent to vertices v and w, which are already colored. This yields two forbidden colors for x. Moreover, if $c(v_1) = c(v_2)$ and/or $c(w_1) = c(w_2)$, this prohibits two more colors for x. Altogether, x has at least six free colors. Next, note that $|Y_i| \leq 3$ for each $i \in [2]$, which together with colors $c(y_1)$ and $c(y_2)$ adds up to at most $2 \times 1 + 2 = 4$ forbidden colors for y. However, if $c(w_1) = c(w_2)$, then, together with the color of w, which is adjacent to y, this possibly yields two more forbidden colors for y. Thus, y has at least four free colors. On the other hand, $|U_i| \leq 4$, for each $i \in [2]$, which together with colors $c(u_1)$ and $c(u_2)$ adds up to at most $2 \times 2 + 2 = 6$ forbidden colors for u. Finally, note that $|Z_i| \leq 4$ for each $i \in [2]$, which together with colors $c(z_1)$ and $c(z_2)$ adds up to at most $2 \times 2 + 2 = 6$ forbidden colors for z. However, if $c(v_1) = c(v_2)$, then, together with the color of v, which is adjacent to u, this possibly yields two more forbidden colors for u. Thus, u has at least two free colors. Finally, note that $|Z_i| \leq 4$ for each $i \in [2]$, which together with colors $c(z_1)$ and $c(z_2)$ adds up to at most $2 \times 2 + 2 = 6$ forbidden colors for z. Therefore, z has at least four free colors.

We color the vertices x, y, z, u in the following way. Firstly, we can color the vertices y and u with different colors in any order since they both have at least two free colors. (Note that we can also have $c(y) = c(v)$ and $c(u) = c(w)$.) This influences the number of free colors for x and z that can drop by two, having four and two free colors, respectively. Finally, we color the remaining vertices x and z, again in an arbitrary order since they both have enough free colors. Since $c(z) \neq c(y)$ and $c(z) \neq c(u)$ (even if $c(y) = c(v)$ and $c(u) = c(w)$) we obtain a 2-frugal coloring of G, which is a contradiction.

Secondly, suppose that there is a vertex y' in G such that y, z, y' induce a triangle in G. Consider $G'' = G - (V(W) \cup \{y'\})$. Let $N_G(y) \cap V(G'') = \{y_1\}$, $N_G(z) \cap V(G'') = \{z_1\}$, $N_G(u) \cap V(G'') = \{u_1, u_2\}$, $N_G(v) \cap V(G'') = \{v_1, v_2\}$, $N_G(w) \cap V(G'') = \{w_1, w_2\}$, and let $N_G(y') \cap V(G'') = \{y'_1, y'_2, y'_3\}$. We further set $Y_1 = N_G(y_1) \setminus \{y\}$, $Z_1 = N_G(z_1) \setminus \{z\}$, $U_i = N_G(u_i) \setminus \{u\}$, $V_i = N_G(v_i) \setminus \{v\}$ and $W_i = N_G(w_i) \setminus \{w\}$ for all $i \in [2]$, and $Y'_i = N_G(y'_i) \setminus \{y'\}$ for all $i \in [3]$. Now let G' be obtained from G'' by adding edges $y'_1y'_2$, u_1u_2, v_1v_2 and w_1w_2, if necessary (see Figure 10(b)). Since G' has smaller order than G and $\Delta(G') \leq 5$, there exists a 2-frugal coloring c of G' using 10 colors.

Now we count the numbers of free colors at each of $V(G) \setminus V(G')$. Note that x has no forbidden color at this moment. Since $|Y'_i| \leq 4$, together with colors $c(y'_i)$, $i \in [3]$, we have at most $3 \times 2 + 3 = 9$ forbidden colors at y', i.e., there is at least one free color at y'. For vertices y and z, $|Y_1| \leq 4$ as well as $|Z_1| \leq 4$, which together with color $c(y_1)$, or $c(z_1)$, imply potential 3 forbidden colors at y, or at z, respectively. In addition, the possibility that $c(y'_i) = c(y'_3)$ for some $i \in [2]$ yields another forbidden color for y and z. This means that there are at least 6 free colors at each of y and z. For vertices u, v, w, each of U_i, V_i and W_i, $i \in [2]$, has at most 4 vertices, which together with colors $c(u_i)$, $c(v_i)$ or $c(w_i)$, imply at most 6 forbidden colors at each of u, v, w, respectively. Hence there are at least 4 free colors at each of u, v, w.

We extend the coloring c of G' to G in the following way. We color y' with a free color, which may decrease the numbers of free colors at x, y, z, u and w by one. Next we color w with

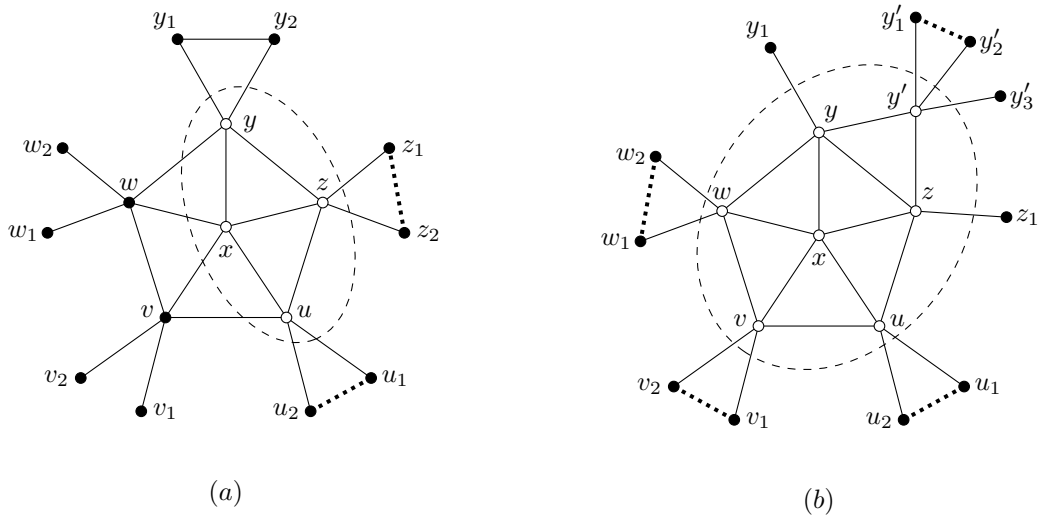


Figure 10: Two cases in the proof of Claim 10. Dashed circle describes the removal of the subgraph, while dotted edges are added in G' (if they were not already there in G); black vertices may have additional neighbors with respect to the maximum degree.

a free color, which may decrease the numbers of free colors at x, y, z, u and v by one. Now we color v with a free color (there are still at least three such colors), which may decrease the numbers of free colors at x, y, z and u by one. At this moment, u has still at least one free color, which we use for coloring u. This may decrease the numbers of free colors at x, y and z by one. For vertices y and z, we have at least two free colors (6 initially minus 4 colors used on vertices y', w, v, u). Then we can clearly color vertices y and z with distinct colors from their free color lists. It remains to color vertex x, which still has at least four free colors (from initial 10 free colors, additional 6 colors may become forbidden for x while coloring vertices y', w, v, u, y and z). Thus, x can be colored with one of the free colors, yielding a 2-frugal coloring of G, a contradiction. □

Claim 11. *If there is a triangle with a vertex of degree* 4 *in* G*, then each of the other two vertices of this triangle lies in at most two triangles.*

Proof. Let x, y, z be the vertices of a triangle T in G. Due to Claim 4, we may assume that $\deg_G(x) = 4$ and $\deg_G(y) = \deg_G(z) = 5$. Suppose to the contrary that z lies in two more triangles. By Claim 5 and 6, z and y lie in only one common triangle, namely T, and the same holds for z and x. Therefore the additional two triangles in which z lies must share an edge for otherwise $\deg_G(z) \geq 6$ which is impossible. Consider $G'' = G - T$, and let $N_G(x) \cap V(G'') = \{x_1, x_2\}$, $N_G(y) \cap V(G'') = \{y_1, y_2, y_3\}$, and $N_G(z) \cap V(G'') = \{z_1, z_2, z_3\}$, where $z_1z_2, z_2z_3 \in E(G)$. Further let $X_i = N_G(x_i) \setminus \{x\}$ for all $i \in [2]$, and $Y_i = N_G(y_i) \setminus \{y\}$ for all $i \in [3]$. Letting $W = \{z_1, z_2, z_3\}$ we define $Z_i = N_G(z_i) \setminus W$ for all $i \in [3]$. Now let G' be obtained from G'' by adding the edges x_1x_2, y_1y_2 and z_1z_3 if necessary (see Fig. 11). Since G' has smaller order than G and $\Delta(G') \leq 5$, there exists a 2-frugal coloring c of G' using 10 colors.

Now, count the number of free colors at vertices x, y and z in G. Note that $|X_i| \leq 4$ for each $i \in [2]$, which together with colors $c(x_1)$ and $c(x_2)$ adds up to at most $2 \times 2 + 2 = 6$

forbidden colors at x. It can happen that either $c(y_1) = c(y_3)$ or $c(y_2) = c(y_3)$, which results in another forbidden color at x. Hence, there are at least three free colors at x. Next, note that $|Y_i| \leq 4$ for each $i \in [3]$, which together with colors $c(y_1)$, $c(y_2)$, and $c(y_3)$ adds up to at most $3 \times 2 + 3 = 9$ forbidden colors. Thus, there is at least one free color at y. Finally, note that $|Z_i| \leq 3$ for each $i \in [3]$, and at most one color can be used twice on each Z_i. Together with colors $c(z_1)$, $c(z_2)$, and $c(z_3)$ this yields $3 \times 1 + 3 = 6$ forbidden colors. Similarly as above note that either $c(y_1) = c(y_3)$ or $c(y_2) = c(y_3)$ can result in another forbidden color for z. Thus, there are at least three free colors at z.

Color the vertices of the triangle T in the following way. First, color y with a free color. Consequently the number of free colors at vertices x and z can drop by one, resulting in at least two free colors at both vertices. In the next two steps, color the vertices x and z with different colors in any order. This gives a 2-frugal coloring of G, which is a contradiction. □

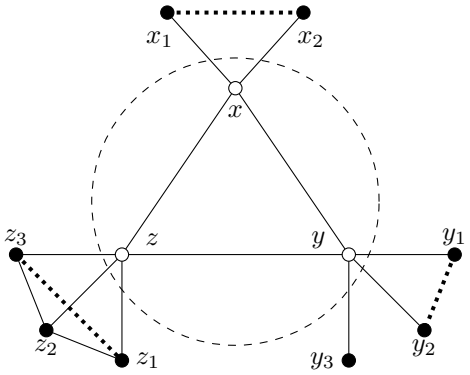


Figure 11: The situation in Claim 11. Dashed circle describes the removal of the triangle, while dotted edges are added in G' (if they are not already there in G); black vertices may have additional neighbors with respect to the maximum degree.

Now, we proceed with the proof of Theorem 1.4. Letting $F(G)$ be the set of faces of G, we assign the initial charge $\mu(z)$ for each element $z \in V(G) \cup F(G)$ to be $\deg_G(z) - 4$ (note that the degree of a face is defined as the number of its boundary edges (or vertices)). By Euler's formula the initial total charge of the graph, which is the sum of all charges, equals

$$\sum_{v \in V(G)} (\deg_G(v) - 4) + \sum_{f \in F(G)} (\deg_G(f) - 4) = 2|E(G)| - 4|V(G)| + 2|E(G)| - 4|F(G)| = -8.$$

Note that while the total charge is negative, only 3-faces contribute to the negative values in the sum, since by Claim 3, $\delta(G) \geq 4$. The following discharging rules are used:

(R1) If v is a vertex of degree 5 that belongs to only one 3-face, then v passes the charge of 1 to that 3-face.

(R2) If v is a vertex of degree 5 that belongs to two 3-faces, then v passes the charge of $\frac{1}{2}$ to the two 3-faces to which it belongs.

(R3) If v is a vertex of degree 5 that belongs to three 3-faces, then v necessarily belongs to a diamond D such that v has degree 3 with respect to D.

(R3.1) If v has degree 3 with respect to only one diamond, then, we let v pass the charge of $\frac{1}{4}$ to each of the two 3-faces that form a diamond and the charge $\frac{1}{2}$ to the remaining 3-face to which it belongs.

(R3.2) If v has degree 3 with respect to two diamonds D, then these diamonds overlap in a 3-face (the "middle" 3-face), and we let v pass the charge of $\frac{1}{2}$ to the middle 3-face and the charge of $\frac{1}{4}$ to each of the remaining two 3-faces.

(R4) If v is a vertex of degree 5 that belongs to five 3-faces, then v is the center of a 5-wheel. By Claim 10, each of its neighbors $v_1, \ldots, v_5$ lies in exactly two triangles (of the same wheel). By Claim 5, each v_i is a vertex of degree 5. Then, we let each v_i pass the charge of $\frac{1}{2}$ to the two 3-faces to which it belongs (in accordance to (R2)).

Let T be a 3-face in G with vertices x, y and z. There are two possibilities depending on whether T contains a vertex of degree 4.

We first consider the case when T contains a vertex of degree 4, say $\deg_G(z) = 4$. Note that T is the unique triangle to which z belongs (by Claim 7), and the remaining two vertices x, y of T have degree 5 (by Claim 4). By Claim 11, x (respectively y) belongs to at most one additional triangle in G, and so by the rules (R1) and (R2), we let x and y each pass the charge of at least $\frac{1}{2}$ to the face T (and the charge of $\frac{1}{2}$ to another 3-face of G if applicable).

For the rest of the proof, we may assume that all vertices of T have degree 5. First, if T lies in a 5-wheel, we derive by (R4) that T receives the charge $\frac{1}{2}$ from two of its vertices, which increases the charge of T by 1. Thus, we may further restrict to the case when T is not part of 5-wheel. Note that, by the rules, each of the vertices of T passes the charge of at least $\frac{1}{4}$ to the 3-face T. Now, if one of the vertices in T lies in at most two triangles, then due to the first two rules it passes the charge of at least $\frac{1}{2}$ to T, which increases the charge of T by at least 1.

Now, we are left with the case in which each of the three vertices of T lies in exactly three 3-faces (four 3-faces is not possible by Claim 9, and five 3-faces is not possible since no vertex of T is the center of a 5-wheel). First, assume that T itself is not part of a diamond. Let x be a vertex in T. Note that x lies in at least one diamond, however x has degree 3 in exactly one diamond D. Hence, (R3.1) is applicable for x, which thus passes the charge of $\frac{1}{2}$ to T, and consequently the charge of T increases by at least 1.

Finally, assume that T is part of a diamond. First assume that there exist two 3-faces that share an edge with T, and thus form two diamonds with it; say diamond D_1 shares an edge with xy and diamond D_2 shares an edge with xz. Then, by using the (R3.2), we infer that x passes the charge of $\frac{1}{2}$ to T, and consequently the charge of T increases by at least 1. The second and remaining case is that exactly one 3-face shares an edge with T, and forms one diamond with it; say this edge is yz. Since x lies in three 3-faces, where the remaining two faces do not share an edge with T, we infer that these two faces share an edge and form a diamond. Applying (R3.1) we infer that x passes the charge of $\frac{1}{2}$ to T, and consequently the charge of T increases by at least 1.

All in all, we derive that T received the charge of at least 1 from its incident vertices by applying the above discharging rules. Therefore, the final total charge of the graph is non-negative, which is a contradiction with the initial charge being -8. The proof is complete.

4 Concluding remarks

The next step in the context of this paper would be to resolve Problem 1.2 for the remaining two maximum degrees, that is, $\Delta \in \{6, 7\}$. We think this may be even more involved than the case $\Delta = 5$. In the light of Conjecture 1.1, the following problem is also natural.

Problem 4.1. *Prove that for every planar graph G with maximum degree $\Delta \geq 8$,*

$$\chi_2^f(G) \leq \left\lfloor \frac{3\Delta}{2} \right\rfloor + 1.$$

Acknowledgements

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